

Page 2: Requirement Analysis Phase

1. Introduction

The **Requirement Analysis Phase** defines the technical, functional, operational, and non-functional specifications necessary to build **Smart Budget AI**. This phase serves as the foundation for software architecture, dataset collection, model selection, and platform scalability.

2. Functional Requirements

2.1 User Management & Onboarding

- User registration, secure login, and multi-factor authentication (MFA).
- Profile setup including currency selection, income source input, fixed monthly bills, and financial goals.

2.2 Data Ingestion & Expense Tracking

- **Automated Parsing:** OCR (Optical Character Recognition) engine to scan physical receipts and extract vendor, date, line items, and total amount.
- **SMS & Bank Statement Parser:** Secure processing of financial notifications and CSV/PDF bank statement uploads.
- **Manual Entry:** Simplified fallback interface for quick offline transaction entry.

2.3 AI & Machine Learning Features

- **Automated Categorization Model:** NLP/Classification algorithm to categorize transactions with high accuracy.
- **Predictive Expense Forecasting:** Time-series analysis model to forecast month-end spending based on current patterns.
- **Smart Alert Engine:** Real-time triggers for overspending, duplicate charges, recurring subscription renewals, and unusual transactions.
- **Conversational AI Assistant:** LLM-powered chatbot to respond to natural language queries regarding account status and financial advice.

2.4 Goal Management & Analytics

- Interactive dashboard with visual charts (pie charts, line graphs, progress bars).
- Custom saving goal tracker with automated daily/weekly micro-saving calculations.

3. Non-Functional Requirements

- **Security & Data Privacy:** End-to-end encryption (AES-256 for data at rest, TLS 1.3 for data in transit), strictly adhering to GDPR/data protection regulations.

- **Performance & Speed:** Real-time transaction processing with AI response times under 2 seconds.
- **Scalability:** Cloud-native architecture capable of supporting concurrent user traffic and growing transaction history.
- **Usability & Accessibility:** Clean, intuitive UI optimized for both mobile and web platforms.
- **Availability:** 99.9% application uptime using cloud infrastructure.

4. Hardware & Software Requirements

Hardware / Cloud Infrastructure

- **Development Workstation:** Core i7/Ryzen 7 Processor, 16GB+ RAM, GPU for machine learning model training.
- **Cloud Platform:** AWS / Google Cloud / Azure instance with scalable database and GPU compute capabilities.

Software Stack

- **Frontend:** React Native / Flutter (Mobile) or React.js (Web).
- **Backend Framework:** Python (FastAPI / Django) or Node.js.
- **Database:** PostgreSQL (Relational Data), Redis (In-Memory Caching), MongoDB (Unstructured Logs).
- **AI/ML Tools:** Python, TensorFlow / PyTorch, Scikit-learn, Tesseract OCR / EasyOCR, OpenAI API / Hugging Face Transformers.

5. Risk Analysis & Mitigation

Risk	Impact	Mitigation Strategy
Data Privacy / Breach Concerns	High	Implement strict data encryption, zero-knowledge storage, and multi-factor authentication.
Inaccurate AI Expense Categorization	Medium	Implement user feedback loops allowing users to manually correct categories and retrain models.
Bank API Integration Limits	Medium	Support OCR receipt scanning and CSV statement imports as reliable backup methods.